

RES-NUM — Numerical Methods, Accuracy and Stability

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1 Domain Purpose and Scope

2 Domain Deliverables

3 RES-NUM-FRM — Numerical Formulation and Discrete Conservation Framework

3.1 Purpose and Scope

This Deliverable records the breadth-first numerical-method decision for the regional tsunami solver. It selects the numerical family used for the two-dimensional nonlinear shallow-water model and records the candidate families that remain useful as references, deferred ideas, or screened alternatives.

The original family-level decision selected finite volume for the regional model. The 2026-07-07 regional gate extends that decision to a research-specification-level numerical model for the regional 2D NLSWE solver. The local three-dimensional CFD family is also finite volume, but it is documented separately from the regional NLSWE discretisation because it solves a collocated, polyhedral, pressure-corrected URANS–VOF system.

3.2 Research Questions

The current phase answers the following questions:

1. Which numerical family is selected for the production regional nonlinear shallow-water solver?
2. Which candidate families are retained as references or deferred concepts?
3. Which lower-level choices remain unresolved after the family-level decision?

3.3 Terminology and Definitions

Four decision statuses are used:

selected Adopted for the current project branch.

provisional Adopted as a working direction pending lower-level comparison or verification.

excluded Removed from the active branch for project-specific reasons.

unresolved Not yet selected, excluded, or formally deferred.

3.4 Literature Evidence

3.4.1 Finite-Difference Method

3.4.1.1 Methodological basis.

Casulli and Cheng [1] developed a semi-implicit finite difference architecture for shallow-water flow in which the free-surface unknowns are reduced to symmetric positive-definite linear systems. The paper is retained for selective implicit treatment, conjugate-gradient and multi-grid compatibility, stable wetting and drying, and the distinction between stability-limited and accuracy-limited time steps.

3.4.1.2 Contextual validation.

Wei et al. [2] used MOST to model the 2011 Japan tsunami and demonstrated a finite-difference family in a near-field tsunami forecast setting. This validates a production tsunami use case for the family, not the exact Casulli–Cheng semi-implicit formulation.

3.4.2 Finite-Volume Method

3.4.2.1 Methodological basis.

Kurganov and Petrova [3] presented a second-order central-upwind method for the Saint-Venant system that simultaneously targets well-balancing and positivity preservation. The method is retained as primary evidence for the regional need to preserve lake-at-rest balance, non-negative depth, discontinuities, and dry or nearly dry states.

3.4.2.2 Contextual validation.

Dutykh, Poncet, and Dias [4] documented the VOLNA tsunami code using finite volumes on unstructured triangular meshes for tsunami generation, propagation, run-up and inundation. The paper is retained as contextual evidence that a finite-volume shallow-water solver can support complex coastal domains, wetting and drying, and tsunami-scale validation cases.

3.4.2.3 Local 3D finite-volume and pressure-correction evidence.

Hirt and Nichols [5] established the VOF representation of moving free boundaries. Vukčević et al. [6] and Roenby, Bredmose, and Jasak [7] support sharp and geometric VOF treatments on finite-volume meshes. Issa [8] supports pressure-correction by operator splitting for incompressible flow. Source roles: ‘3D-VOF’ and ‘3D-FVM’. Project conclusion: the local solver shall use conservative finite-volume transport for mass, momentum and phase fraction, with pressure correction enforcing incompressibility. Limitation: these sources establish the numerical family and pressure-correction principle, not the final linear solver, preconditioner or mesh-convergence tolerance.

3.4.3 Lattice/Discrete Boltzmann Method

3.4.3.1 Methodological basis.

Meng et al. [9] derived discrete Boltzmann models for shallow-water equations using polynomial equilibria. The paper is retained for collision-level conservation, local stream–collision structure, GPU-relevant locality, and the conditional ability to improve supercritical capability through higher-order expansion and quadrature.

3.4.3.2 Contextual validation gap.

No tsunami-context validation paper for the lattice or discrete Boltzmann branch has yet passed the project evidence gate. The reviewed paper does not establish genuinely dry wetting and drying, fully discrete well-balancing, positivity of depth in the project setting, or adequate treatment of irregular coastlines and bathymetry.

3.4.4 Spectral-Method Screening

No detailed spectral-method literature-evidence subsection is created for the current comparison. Spectral and pseudospectral methods are screened at the project-fit level owing to non-periodic irregular coastal boundaries, wetting and drying, bores and hydraulic discontinuities, complex bathymetry, local refinement requirements, and the loss of the principal global-spectral advantages once extensive boundary treatment, filtering, and domain decomposition are required.

3.5 Governing Relations and Quantitative Definitions

3.6 Discrete Formulation

The regional production solver is selected as a cell-centred finite-volume method on unstructured triangular or polygonal control volumes. This selection concerns the spatial conservation framework, not the full lower-level numerical method.

Spatial discretisation and source-term integration are separate decisions. The regional non-linear shallow-water equations already include bed slope, bottom friction, Coriolis forcing, earthquake or moving-bottom forcing where activated, and wetting–drying requirements. The provisional source-term treatment is:

bed slope coupled well-balanced hydrostatic reconstruction;

bottom friction semi-implicit update;

Coriolis exact or structure-preserving rotational update.

The detailed derivations and notation for friction and Coriolis remain owned by the mathematical-model Deliverables. This numerical Deliverable records how those terms are integrated within the selected finite-volume branch.

3.6.1 Local 3D Collocated Polyhedral Finite Volume

The local URANS–VOF solver uses cell-centred collocated fields on polyhedral control volumes. For a local cell K_i with volume V_i , faces f , outward area vectors $\mathbf{S}_f$ and neighbouring cells $\mathcal{N}(i)$, a transported scalar or vector component ψ is stored as:

$$\psi_i(t) = \frac{1}{V_i} \int_{K_i} \psi(\mathbf{x}, t) dV \quad (3.1)$$

The face-normal volume flux is:

$$\phi_f = \mathbf{u}_f \cdot \mathbf{S}_f \quad (3.2)$$

The conservative semi-discrete transport form is:

$$\frac{d}{dt} (V_i \psi_i) = - \sum_{f \in \partial K_i} \phi_f \psi_f + \sum_{f \in \partial K_i} \Gamma_f \nabla \psi_f \cdot \mathbf{S}_f + V_i S_{\psi,i} \quad (3.3)$$

After temporal and spatial linearisation each transported field is assembled into an algebraic cell equation:

$$a_{P,i} \psi_i + \sum_{j \in \mathcal{N}(i)} a_{N,ij} \psi_j = b_i \quad (3.4)$$

Source terms may be linearised as:

$$S_{\psi,i} = S_{C,i} + S_{P,i} \psi_i \quad (3.5)$$

with coefficient updates:

$$a_{P,i}^{\text{new}} = a_{P,i}^{\text{base}} - V_i S_{P,i} \quad (3.6)$$

$$b_i^{\text{new}} = b_i^{\text{base}} + V_i S_{C,i} \quad (3.7)$$

Pressure correction enforces the discrete incompressibility constraint:

$$\sum_{f \in \partial K_i} \phi_f^{n+1} = 0 \quad (3.8)$$

The provisional pressure-correction form is:

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \mathbf{D}\nabla p' \quad (3.9)$$

$$p^{n+1} = p^* + p' \quad (3.10)$$

where $\mathbf{u}^*$ is the momentum-predictor velocity, p' is the pressure correction and $\mathbf{D}$ is the discrete velocity-correction operator implied by the momentum matrix. Rhie–Chow or an equivalent pressure-consistent face-flux interpolation shall be used on collocated meshes if required to prevent pressure–velocity decoupling.

3.7 Conservation Properties

3.8 Consistency and Formal Accuracy

3.9 Stability Requirements

3.10 Mesh and Time-Step Requirements

3.11 Numerical Failure Modes

3.12 Verification Requirements

3.13 Candidate Approaches

The candidate regional numerical families considered in this phase are:

- cell-centred finite volume;
- finite difference;
- lattice or discrete Boltzmann;
- spectral or pseudospectral methods.

3.14 Critical Comparison

3.14.1 Comparative Assessment

The comparison is project-specific rather than a universal ranking of numerical analysis families.

Conservation Finite volume is strongest for direct conservation of water depth and horizontal momentum through face fluxes. Finite difference remains useful but requires careful conservative formulation. Lattice/discrete Boltzmann has strong collision-level conservation, while spectral conservation depends on the formulation and filtering.

Well-balancing The selected finite-volume branch has direct support through hydrostatic reconstruction and well-balanced shallow- water fluxes. Finite-difference and Boltzmann branches retain useful ideas but require further project-specific proof.

Positivity Finite volume provides the clearest route to non-negative water depth in wetting–drying regions. The reviewed Boltzmann paper does not establish the required dry-front robustness.

Wetting and drying VOLNA and the central-upwind evidence support finite volume for inundation. Finite difference remains credible as a reference family through MOST and semi-implicit shallow-water literature.

Bore and shock treatment Finite volume is selected for weak solutions, bores, hydraulic discontinuities, and shoreline Riemann treatments.

Coastline and mesh flexibility Unstructured finite volumes match irregular coastlines, harbour geometry, local mesh grading, and face-based boundary patches.

Phase and amplitude accuracy The exact regional flux, reconstruction, timestep and optional dispersive extension remain unresolved and shall be tested against propagation and inundation benchmarks.

Time-step and linear-system structure Finite difference is retained for selective implicitness, reduced free-surface systems, symmetric positive-definite solves, conjugate-gradient compatibility and multigrid compatibility.

Parallel and GPU structure Finite volume and lattice/discrete Boltzmann both expose local operations. Lattice/discrete Boltzmann remains deferred until the tsunami-validation and wet/dry evidence gaps are closed.

Coupling compatibility Finite volume produces conservative face fluxes suitable for the two-dimensional–three-dimensional interface and shares cell, face, flux and boundary-patch abstractions with the OpenFOAM-inspired local solver architecture.

3.15 Assumptions and Applicability

3.16 Limitations and Failure Conditions

The finite-volume selection does not imply that finite difference, lattice/discrete Boltzmann, or spectral methods are inferior in general. The screening is limited to the current regional tsunami solver branch.

Finite difference is not selected for production regional execution but is retained as a numerical reference and as a source of semi-implicit architectural ideas. Lattice/discrete Boltzmann is removed from the active regional shortlist until tsunami validation, positivity, well-balancing, dry wetting and drying, and complex-bathymetry evidence are available. Spectral methods are screened out of the active regional comparison owing to project-fit constraints rather than mathematical invalidity.

3.17 Project-Specific Conclusions

The regional nonlinear shallow-water solver shall use a cell-centred finite-volume method. The primary rationale is physical and numerical: direct conservation, weak-solution compatibility, well-balanced bathymetry, positivity-preserving wetting and drying, unstructured coastline representation, local mesh grading, and conservative two-dimensional–three-dimensional interface fluxes.

OpenFOAM compatibility is a secondary architectural benefit. The regional solver is not required to be implemented inside OpenFOAM.

The local three-dimensional solver shall also use finite volume, but with collocated polyhedral control volumes, VOF phase transport and pressure correction for incompressibility. OpenFOAM is retained as the principal architectural reference for this local branch.

3.18 Selected Approach and Rationale

3.18.1 Regional Solver Decision

The selected regional numerical family is:

cell-centred finite-volume method

The selected regional solver composition is:

- unstructured triangular or polygonal control volumes;
- second-order MUSCL reconstruction;
- weighted least-squares gradients;
- Barth–Jespersen limiting;
- well-balanced hydrostatic reconstruction;
- positivity-preserving wet/dry treatment;
- HLLC primary flux with HLL/HLLE fallback near dry or unsafe states;
- SSPRK(3,3) transport;
- Strang-split semi-implicit Manning friction;
- characteristic or radiation open boundaries;
- sponge damping for artificial ocean cuts where required;
- mesh refinement driven by bathymetry, shoreline, source, structures, wet/dry fronts and validation observables.

Dry-depth tolerances, exact wave-speed estimates, final mesh sizes and domain-specific sponge dimensions remain verification or implementation parameters, not open research-family decisions.

3.18.2 Retained FDM Concepts

Finite difference is retained for:

- selective implicit treatment;
- reduced free-surface systems;
- symmetric positive-definite linear systems;
- conjugate-gradient and multigrid compatibility;
- stable wet/dry face handling;
- the distinction between stability-limited and accuracy-limited timesteps.

3.18.3 Excluded and Deferred Alternatives

Lattice/discrete Boltzmann is removed from the active regional shortlist. It remains a deferred research candidate if tsunami-context validation, robust dry wetting and drying, fully discrete well-balancing, and irregular-coastline treatment are demonstrated.

Spectral and pseudospectral methods are screened out of the active regional solver comparison for this project branch.

3.18.4 Local 3D Solver Decision

The selected local numerical family is:

cell-centred collocated polyhedral finite volume with bounded VOF phase transport and pressure correction for incompressible two-phase URANS

The selected baseline turbulence closure for this finite-volume branch is k - ω SST. LES remains a higher-fidelity comparison mode, and particle, finite-element and staggered-grid methods remain references or specialist alternatives rather than the production local hydrodynamic family.

3.19 Unresolved Decisions

The following regional implementation and validation parameters remain open:

- dry-depth tolerance;
- exact HLL/HLLC wave-speed safeguards;
- final mesh refinement levels;
- final CFL value after positivity and wet/dry verification;

-
- optional dispersive extension and its activation gate;
 - domain-specific open-boundary and sponge placement;
 - coupling residuals and convergence criteria for later multidimensional coupling.
 - exact local VOF advection and compression scheme;
 - exact pressure-correction sequence and pressure-consistent face-flux interpolation;
 - local wall-function and rough-wall treatment;
 - local linear solvers, preconditioners and residual tolerances.

3.20 Requirements and Handoffs

The selected regional solver family hands off the following requirements:

- ‘RES-NUM-MSH’ shall define unstructured mesh, local grading and adaptive-mesh strategy;
- ‘RES-NUM-FLX’ shall compare HLL, HLLC, central-upwind and related shoreline-aware fluxes;
- ‘RES-NUM-SRC’ shall formalise bed-slope, friction and Coriolis integration;
- ‘RES-NUM-TIM’ shall select the SSP-RK order and any semi-implicit split;
- ‘RES-NUM-WET’ shall select wet/dry thresholds and positivity controls;
- ‘RES-MOD-MUL’ and later Software work shall use conservative face-flux outputs for the two-dimensional–three-dimensional interface.
- ‘RES-NUM-SPEC’ shall define the local URANS–VOF solve order, pressure-correction loop, residual checks and timestep rejection rules.

3.21 Deliverable Completion Record

4 RES-NUM-MSH — Mesh Topology, Resolution and Geometric Discretisation

4.1 Purpose and Scope

4.2 Research Questions

4.3 Terminology and Definitions

4.4 Literature Evidence

Roenby, Bredmose, and Jasak [7] and Marić, Marschall, and Bothe [10] support geometric VOF advection on general or unstructured finite-volume meshes. Source role: ‘3D-VOF’. Project conclusion: the local 3D mesh may use polyhedral finite volumes and refine around the air–water interface and barrier interaction region. Limitation: the sources do not set the final project cell sizes, refinement ratios, wall-normal spacing or adaptive-mesh triggers.

4.5 Governing Relations and Quantitative Definitions

4.6 Discrete Formulation

The regional finite-volume mesh is specified as a set of straight-edged polygonal control volumes. The practical production mesh is unstructured triangular; quadrilateral meshes are retained for canonical tests and simple offshore corridors. The cell update is shape-independent and uses only the geometry in Equation (12.4).

The local finite-volume mesh is specified as a three-dimensional polyhedral mesh with cell volumes, face area vectors, owner–neighbour connectivity, boundary patches and wall-distance information. The local mesh shall represent the fixed terrain and barrier surfaces while allowing refinement near the free surface and local impact region.

4.7 Conservation Properties

4.8 Consistency and Formal Accuracy

4.9 Stability Requirements

4.10 Mesh and Time-Step Requirements

Mesh quality directly affects gradient reconstruction, limiter activation, hydrostatic source balance, wet/dry stability and the CFL timestep in Equation (12.36). The mesh shall therefore

record cell area, face length, outward normal, face centroid, cell centroid and neighbour connectivity for every regional cell.

For the local three-dimensional URANS–VOF model, refinement shall be considered in the following zones:

- incident-wave inlet and relaxation region;
- expected free-surface path and interface-fragmentation region;
- run-up and overtopping paths;
- barrier leading edges, corners, crest and openings;
- near-wall terrain and barrier boundary layers;
- pressure-probe and force-integration surfaces;
- reflected and transmitted wave measurement locations;
- downstream wake and recirculation region;
- outlet and side-boundary absorption regions.

The local mesh-quality record shall include non-orthogonality, skewness, aspect ratio, local cell volume, face area, wall distance, local Courant number, interface Courant number and wall-function validity indicators. These quantities are numerical-quality outputs rather than case-study conclusions.

4.11 Numerical Failure Modes

4.12 Verification Requirements

4.13 Candidate Approaches

4.14 Critical Comparison

4.15 Assumptions and Applicability

4.16 Limitations and Failure Conditions

4.17 Project-Specific Conclusions

4.18 Selected Approach and Rationale

The selected regional mesh approach is:

- general polygonal finite-volume equations;

-
- unstructured triangles as the primary case-study mesh;
 - quadrilaterals for verification, simple corridors and structured comparisons;
 - refinement around bathymetric gradients, shorelines, wet/dry fronts, source regions, candidate defences, gauge/observation locations and regional-to-local extraction boundaries.

The selected local mesh approach is:

- body-fitted or otherwise geometry-resolving polyhedral finite-volume cells for the fixed terrain and fixed barrier;
- stable boundary patches matching the local boundary-role partition defined in ‘RES-MOD-IBC’;
- local refinement around the free surface, impact surfaces, overtopping region, wake and validation probes;
- wall-normal resolution sufficient for the selected high-Re wall-function treatment;
- mesh-convergence studies based on free-surface, pressure, force, moment and overtopping metrics.

4.19 Unresolved Decisions

The mesh topology is specified at research level. Final mesh sizes, quality thresholds, refinement ratios and QGIS-derived domain extents remain open until ‘RES-DAT-GEO’, ‘RES-GEO-REP’ and validation planning close the case-study geometry.

For the local model, final wall-normal spacing, roughness treatment, static-versus-adaptive refinement policy, interface-refinement thresholds and domain extents remain open until the case-study defence geometry and validation data are locked.

4.20 Requirements and Handoffs

‘RES-NUM-ERR’ shall record mesh-quality and refinement indicators for local simulations. ‘RES-VER-CNV’ shall define local mesh-convergence tests for free-surface elevation, pressure, force, moment, overtopping volume and energy metrics.

4.21 Deliverable Completion Record

5 RES-NUM-FLX — Spatial Reconstruction and Numerical Flux Selection

5.1 Purpose and Scope

5.2 Research Questions

5.3 Terminology and Definitions

5.4 Literature Evidence

5.5 Governing Relations and Quantitative Definitions

5.6 Discrete Formulation

The regional reconstruction state is $\mathbf{W} = [\eta, u, v, b]^T$. The selected reconstruction is MUSCL piecewise-linear reconstruction using weighted least-squares gradients and a Barth–Jespersen limiter, as defined in Equations (12.15), (12.16) and (12.20). Reconstruction reverts to first order near wet/dry fronts where needed for positivity.

5.7 Conservation Properties

5.8 Consistency and Formal Accuracy

5.9 Stability Requirements

HLLC is selected as the primary flux in wet, valid states. HLL or HLLE is the fallback near dry, degenerate or unsafe states. Face states are rotated into normal/tangential coordinates before the Riemann solve, as recorded in Equations (12.21) and (12.23).

5.10 Mesh and Time-Step Requirements

5.11 Numerical Failure Modes

5.12 Verification Requirements

5.13 Candidate Approaches

5.14 Critical Comparison

5.15 Assumptions and Applicability

5.16 Limitations and Failure Conditions

5.17 Project-Specific Conclusions

5.18 Selected Approach and Rationale

Selected regional flux/reconstruction hierarchy:

- second-order MUSCL reconstruction in smooth wet regions;
- weighted least-squares gradients on polygonal meshes;
- Barth–Jespersen multidimensional limiting;
- hydrostatically reconstructed face states before fluxing;
- HLLC for the normal shallow-water Riemann problem in valid wet states;
- HLL/HLLE fallback for dry, nearly dry or otherwise unsafe states.

5.19 Unresolved Decisions

The flux family, reconstruction family and limiter family are no longer open. Exact HLLC wave-speed estimates, dry-state switching constants and case-specific limiter safeguards remain verification and implementation parameters.

5.20 Requirements and Handoffs

5.21 Deliverable Completion Record

6 RES-NUM-SRC — Numerical Treatment of Source Terms and Well-Balancing

6.1 Purpose and Scope

6.2 Research Questions

6.3 Terminology and Definitions

6.4 Literature Evidence

6.5 Governing Relations and Quantitative Definitions

6.6 Discrete Formulation

The regional bed source is discretised with hydrostatic reconstruction. At each face, the common hydrostatic bed and reconstructed depths are defined by Equations (12.24) to (12.26). The second-order bed-gradient source uses Equation (12.27).

Manning friction is advanced by the split semi-implicit update owned by ‘RES-NUM-TIM’. Sponge damping is represented by Equations (12.28), (12.30) and (12.31).

6.7 Conservation Properties

6.8 Consistency and Formal Accuracy

6.9 Stability Requirements

6.10 Mesh and Time-Step Requirements

6.11 Numerical Failure Modes

6.12 Verification Requirements

The source treatment shall be verified against lake-at-rest states over non-flat bathymetry, wet/dry shoreline motion, Manning-only decay, a moving-bottom generation case, and sponge-layer reflection diagnostics.

6.13 Candidate Approaches

6.14 Critical Comparison

6.15 Assumptions and Applicability

6.16 Limitations and Failure Conditions

6.17 Project-Specific Conclusions

6.18 Selected Approach and Rationale

The selected regional source-term approach is:

- hydrostatic reconstruction for bed-slope balance;
- matching second-order bed-gradient source in cells;
- dynamic bed updates at Runge–Kutta stage times;
- Strang-split Manning friction;
- polynomial sponge ramp with baseline $p = 2$, $A_{\text{target}} = 0.01$ and $L_{\text{sp}} = 50\text{--}100$ km.

6.19 Unresolved Decisions

Final roughness fields, sponge placement, source-time functions and lake-at-rest tolerances remain open until data selection and verification are executed.

6.20 Requirements and Handoffs

6.21 Deliverable Completion Record

7 RES-NUM-TIM — Time Integration, CFL Control and Stiffness Treatment

7.1 Purpose and Scope

7.2 Research Questions

7.3 Terminology and Definitions

7.4 Literature Evidence

7.5 Governing Relations and Quantitative Definitions

7.6 Discrete Formulation

The regional time update is SSPRK(3,3) for the transport/source residual excluding Manning friction, combined with Strang-split semi-implicit Manning friction: Equation (12.34). Full IMEX RK is not selected for the regional baseline because the retained stiff term is cell-local Manning friction.

The local URANS–VOF model uses a segregated transient update. Within each accepted time step the VOF transport, mixture-property update, momentum predictor, pressure correction, turbulence equations and metric extraction are ordered as specified in ‘RES-NUM-SPEC’.

7.7 Conservation Properties

7.8 Consistency and Formal Accuracy

7.9 Stability Requirements

7.10 Mesh and Time-Step Requirements

The regional timestep is controlled by the face signal speed and cell geometry in Equations (12.35) to (12.37). The provisional CFL value is $C_{\text{CFL}} = 0.5$, subject to positivity and wet/dry verification.

The local timestep shall be the minimum of flow-advection, VOF interface, gravity-wave and diffusion constraints:

$$\Delta t_{\text{local}} = \min_i (\Delta t_{\text{CFL},i}, \Delta t_{\alpha,i}, \Delta t_{\text{g},i}, \Delta t_{\nu,i}) \quad (7.1)$$

The local flow Courant constraint is:

$$\Delta t_{\text{CFL},i} = C_{\text{CFL}} \frac{V_i}{\sum_{f \in \partial K_i} |\phi_f| + \epsilon_\phi} \quad (7.2)$$

The VOF/interface Courant constraint is:

$$\Delta t_{\alpha,i} = C_\alpha \frac{V_i}{\sum_{f \in \partial K_i} |\phi_{\alpha,f}| + \epsilon_\phi} \quad (7.3)$$

The local gravity-wave constraint is:

$$\Delta t_{\text{g},i} = C_{\text{g}} \frac{\Delta_i}{\sqrt{gh_i} + \epsilon_{\text{g}}} \quad (7.4)$$

The diffusion constraint is:

$$\Delta t_{\nu,i} = C_\nu \frac{\Delta_i^2}{\nu_{\text{eff},i} + \epsilon_\nu} \quad (7.5)$$

where Δ_i is a representative local cell length and $\nu_{\text{eff},i} = \mu_{\text{eff},i}/\rho_i$. The constants C_{CFL} , C_α , C_{g} and C_ν shall be verified for the selected VOF scheme, pressure-correction sequence, mesh and wall treatment.

7.11 Numerical Failure Modes

7.12 Verification Requirements

7.13 Candidate Approaches

7.14 Critical Comparison

7.15 Assumptions and Applicability

7.16 Limitations and Failure Conditions

7.17 Project-Specific Conclusions

7.18 Selected Approach and Rationale

The selected regional time-integration approach is SSPRK(3,3) transport with Strang-split Manning friction. This preserves an explicit shock-capturing transport update while treating the friction damping in a stable cell-local substep.

The selected local time-control approach is a segregated transient URANS–VOF update with timestep selection governed by Equation (7.1). The final timestep is an accepted-time-step value, not merely a requested value; failed boundedness or convergence checks may force timestep rejection.

7.19 Unresolved Decisions

The timestep method is selected. The final CFL constant, local subcycling policy, checkpoint interval and failure-recovery strategy are implementation/validation decisions.

For the local model, the open timestep decisions are final Courant limits, pressure-corrector count, turbulence subcycling if required, adaptive timestep-reduction factor and coupling-history sampling interval.

7.20 Requirements and Handoffs

7.21 Deliverable Completion Record

8 RES-NUM-WET — Wetting–Drying, Positivity and Discontinuity Treatment

8.1 Purpose and Scope

8.2 Research Questions

8.3 Terminology and Definitions

8.4 Literature Evidence

8.5 Governing Relations and Quantitative Definitions

8.6 Discrete Formulation

The regional model uses positivity-preserving wetting–drying logic and does not use regional VOF. Positivity preservation enforces $h_i \geq 0$; wetting–drying additionally controls shoreline motion, near-dry velocities, reconstruction, flux fallback, draining and dry reset.

The selected dry reset and draining limiter are Equations ([12.32](#)) and ([12.33](#)).

8.7 Conservation Properties

Outgoing donor fluxes shall be scaled consistently so the same limited face flux is applied to both adjacent cells with opposite signs. Dry resets shall be audited as controlled positivity corrections rather than silent conservation-preserving fluxes.

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- 8.8 Consistency and Formal Accuracy**
 - 8.9 Stability Requirements**
 - 8.10 Mesh and Time-Step Requirements**
 - 8.11 Numerical Failure Modes**
 - 8.12 Verification Requirements**
 - 8.13 Candidate Approaches**
 - 8.14 Critical Comparison**
 - 8.15 Assumptions and Applicability**
 - 8.16 Limitations and Failure Conditions**
 - 8.17 Project-Specific Conclusions**
 - 8.18 Selected Approach and Rationale**

The selected regional wet/dry treatment combines hydrostatic reconstruction, HLL/HLLE fall-back, first-order near-dry reconstruction, a draining limiter, velocity regularisation and post-stage dry reset.

8.19 Unresolved Decisions

Final values of h_{dry} , velocity-regularisation depth, near-dry reconstruction threshold and mass-correction audit tolerances remain verification parameters.

8.20 Requirements and Handoffs

8.21 Deliverable Completion Record

9 RES-NUM-IBC — Discrete Initial, Boundary and Interface Treatments

9.1 Purpose and Scope

9.2 Research Questions

9.3 Terminology and Definitions

9.4 Literature Evidence

9.4.1 Takase et al. (2016)

Takase et al. [11] demonstrated the coupling of independently meshed two-dimensional shallow-water and three-dimensional Navier–Stokes domains through multiple-point constraints and transmission relations. The method enforced compatibility of velocity and pressure despite differences in mesh dimensionality and interface node arrangement.

The paper is retained as evidence for the interface requirements rather than as a numerical formulation to reproduce directly. The present project requires an interface treatment compatible with a finite-volume regional solver, independent three-dimensional discretisation, saved restart states, and both one-way and bidirectional execution.

9.4.2 Romano et al. (2025)

Romano et al. [12] transferred a time-dependent three-dimensional tsunami signal into a reduced two-dimensional propagation model and examined the sensitivity of the resulting model chain to the coupling-boundary position. The study showed that transfer accuracy depended on whether the high-fidelity model had already resolved the nonlinear and geometrically controlled transformations that the receiving model could not represent.

The paper is retained as evidence for physics-based interface placement, sequential restart coupling, and preservation of transmitted waveform content. Its free-surface-based, frequency-domain transfer is treated as one case-specific implementation rather than as the interface specification for the present finite-volume shallow-water model.

9.4.3 Higuera et al. (2013) and Jacobsen et al. (2012)

Higuera, Lara, and Losada [13] and Jacobsen, Fuhrman, and Fredsoe [14] are retained as evidence that Navier–Stokes/VOF numerical wave tanks require wave-generation and absorption boundary treatments designed around the intended incident wave signal. Source role: ‘3D-WAVEBC’. Project conclusion: local inlet, outlet and lateral boundary faces shall preserve the

incident history while preventing artificial reflection from contaminating run-up, overtopping and loading outputs. Limitation: exact relaxation profiles and numerical coefficients remain case-study verification choices.

9.5 Governing Relations and Quantitative Definitions

9.6 Discrete Formulation

For the regional solver, boundary conditions are imposed as face states during residual evaluation. Internal and boundary fluxes are:

$$\hat{\mathbf{F}}_f = \hat{\mathbf{F}}(\mathbf{U}_{i,f}, \mathbf{U}_{j,f}, \mathbf{n}_f) \quad (9.1)$$

$$\hat{\mathbf{F}}_f = \hat{\mathbf{F}}(\mathbf{U}_{i,f}, \mathbf{U}_{B,f}, \mathbf{n}_f) \quad (9.2)$$

The baseline regional boundary assignments are:

- offshore artificial boundary: characteristic or radiation open boundary;
- lateral artificial ocean cuts: radiation/open boundary plus sponge damping where required;
- physical coastline: topography plus wetting–drying, not an outer wall boundary;
- true wall or resolved structure: reflective boundary state;
- debug fallback only: transmissive or zero-gradient state.

For a reflective wall, the boundary state preserves depth and reverses normal velocity:

$$h_B = h_I \quad (9.3)$$

$$u_{n,B} = -u_{n,I} \quad (9.4)$$

$$u_{t,B} = u_{t,I} \quad (9.5)$$

The two-dimensional-to-three-dimensional transfer shall construct a local boundary or initial state from:

$$\mathcal{D}_{2 \rightarrow 3} = \{h, \eta, \bar{u}, \bar{v}, h\bar{u}, h\bar{v}\}_\Gamma \quad (9.6)$$

The transferred quantities shall be mapped to:

$$\mathcal{D}_{3,\Gamma} = \{\bar{u}(z), p(z), \alpha(z)\}_\Gamma \quad (9.7)$$

The initial lifting method may use a depth-uniform horizontal velocity profile where the interface lies within a hydrostatic, weakly sheared region. Alternative profiles shall remain available where evidence shows that a uniform reconstruction is inadequate. Hydrostatic pressure may be initialised using:

$$p(z) = \rho g \max(\eta - z, 0) \quad (9.8)$$

The local three-dimensional solver shall subsequently develop the resolved vertical velocity, shear, separation, and vortical structure.

9.6.1 Local Three-Dimensional Boundary-Face States

For each local boundary face f , the discrete boundary state contains:

$$\mathcal{B}_f^{3D} = \{\alpha_f, \rho_f, \mu_f, \mathbf{u}_f, p_f, k_f, \omega_f, \phi_f\} \quad (9.9)$$

At Γ_{in} , the boundary state is reconstructed from the regional interface history using the inlet free-surface, velocity and pressure reconstruction rules defined in ‘RES-MOD-IBC’. The discrete inlet flux is:

$$\phi_{f,\text{in}} = \mathbf{u}_{f,\text{in}} \cdot \mathbf{S}_f \quad (9.10)$$

At Γ_{out} and Γ_{side} , the boundary state shall use an open, radiation or relaxation treatment. The accepted treatment shall report reflected-wave contamination at the measurement locations rather than assuming that a zero-gradient state is non-reflective.

At Γ_{top} , the air boundary shall act as an open atmospheric boundary with a pressure reference compatible with the pressure-correction equation. At Γ_{T} and Γ_{B} , no penetration is imposed by:

$$\mathbf{u}_f \cdot \mathbf{n}_f = 0 \quad (9.11)$$

and the tangential velocity is treated using the selected no-slip or wall-function discretisation. For the high-Reynolds-number baseline, scalable wall-function quantities shall be applied consistently to velocity, k , ω and wall shear on terrain and barrier patches. The exact wall-function constants and roughness parameters remain open until the case-study material and validation data are selected.

The three-dimensional-to-two-dimensional restriction operator shall construct the receiving shallow-water state from the resolved three-dimensional solution.

The depth shall be obtained from the free-surface and bed elevations:

$$h = \eta - z_b \quad (9.12)$$

The depth-integrated discharges shall be obtained from the resolved horizontal velocity field:

$$q_x = \int_{z_b}^{\eta} u \, dz \quad (9.13)$$

$$q_y = \int_{z_b}^{\eta} v \, dz \quad (9.14)$$

The restricted conservative state is therefore:

$$\vec{U}_{2D,\Gamma} = [h, q_x, q_y]^T \quad (9.15)$$

A free-surface history alone shall not be treated as sufficient for the nonlinear shallow-water receiver unless an additional closure supplies a compatible discharge field. The restriction shall retain the spatial variation of the three-dimensional solution along the interface rather than replacing the complete section with one scalar waveform.

9.7 Conservation Properties

The normal volume flux transferred across an interface segment shall satisfy:

$$\int_{\Gamma_{3D}} \vec{u} \cdot \vec{n} \, dA = \int_{\Gamma_{2D}} h \vec{u} \cdot \vec{n} \, ds \quad (9.16)$$

The corresponding momentum transfer shall be monitored using a discrete interface balance:

$$\vec{\varepsilon}_{\text{mom}} = \vec{\Phi}_{\text{mom},3D} - \vec{\Phi}_{\text{mom},2D} \quad (9.17)$$

The exact momentum-flux definition shall be consistent with the governing equations and numerical fluxes used by both solvers.

9.8 Consistency and Formal Accuracy

The interface treatment shall be regarded as a numerical transformation whose error must be measured separately from the errors of either solver.

For any transferred scalar or flux quantity ϕ , the interface error may be represented as:

$$\varepsilon_{\Gamma,\phi} = \phi_{\text{received}} - \mathcal{T}_{\Gamma}(\phi_{\text{donor}}) \quad (9.18)$$

where $\mathcal{T}_{\Gamma}$ denotes the complete spatial and temporal mapping operation.

The mapping shall be assessed for:

- amplitude preservation;
- phase preservation;
- normal mass flux;
- depth-integrated momentum flux;
- spatial variation along the interface;
- relevant wave-frequency content;
- wet and dry extent;
- accumulated interpolation error.

9.9 Stability Requirements

9.10 Mesh and Time-Step Requirements

The interface treatment shall support:

- non-conforming spatial meshes;
- conservative interpolation or projection;
- different two-dimensional and three-dimensional time steps;
- temporal interpolation of incident states;
- configurable coupling intervals;
- subcycling of the local three-dimensional solver;
- synchronisation at each bidirectional exchange point.

For simultaneous coupling, neither solver shall advance beyond a required exchange time until the interface data from the other solver are available. The software implementation may realise this requirement using a task join, future, condition variable, or equivalent synchronisation primitive.

Saved interface data shall include the physical sampling interval, solver time, spatial coordinate system, interpolation method, and all metadata required to reproduce the donor signal deterministically.

The sampling interval Δt_Γ shall satisfy the Nyquist condition for the highest frequency that materially affects the quantities of interest:

$$f_{s,\Gamma} = \frac{1}{\Delta t_\Gamma} \geq 2f_{\max} \quad (9.19)$$

where $f_{\max}$ is the highest physically relevant retained frequency.

Spectral comparison shall be used as a diagnostic even when both solvers operate entirely in the time domain. This comparison shall identify attenuation, amplification, phase distortion, aliasing, or low-pass filtering introduced by checkpoint storage, interpolation, or interface reconstruction.

9.11 Numerical Failure Modes

Potential interface failure modes include:

- loss of mass or momentum during spatial projection;
- artificial reflection at the model boundary;
- suppression of physical reflected waves in one-way mode;
- inconsistent free-surface elevation and pressure;
- nonphysical vertical profiles introduced by the lifting operator;
- temporal phase error caused by interpolation or coupling lag;
- instability caused by explicit bidirectional exchange;
- overspecification of the three-dimensional boundary;
- wetting or drying mismatch across the interface;
- thread or process oversubscription during concurrent comparison runs.

The Romano et al. implementation demonstrates that a case-specific transfer may be successful with free-surface elevation as the prescribed quantity. That choice is not adopted directly for the present nonlinear shallow-water receiver, whose conservative state also requires depth-integrated discharge.

Similarly, the paper’s frequency-domain mild-slope calculation is not used as the primary regional model. Nevertheless, the observed sensitivity to retained frequency content establishes that temporal downsampling or interpolation may remove physically important short-period components even in a time-domain coupling implementation.

Potential failure modes derived from this comparison include:

- transferring into the reduced model before high-fidelity processes have decayed;
- omitting the discharge associated with a transferred free-surface history;
- suppressing spatial variation along a wide interface;
- filtering relevant high-frequency components;
- contaminating the transferred state with an artificial boundary;
- placing the interface within a strong geometric contraction, expansion, or resonance region;
- interpreting agreement at one interface position as evidence of interface independence.

9.12 Verification Requirements

The interface implementation shall be tested using:

- still-water preservation;
- steady uniform-flow transmission;
- solitary-wave transmission;
- propagation from two dimensions into three dimensions;
- propagation from three dimensions back into two dimensions;
- reflection from a controlled obstacle;
- non-conforming mesh transfer;
- time-step and coupling-interval refinement;
- interface-location sensitivity;
- one-way replay against simultaneous coupling.

Each test shall report free-surface error, phase error, reflected-wave error, mass imbalance, momentum imbalance, and the relevant engineering quantities of interest.

9.13 Candidate Approaches

9.14 Critical Comparison

9.15 Assumptions and Applicability

9.16 Limitations and Failure Conditions

9.17 Project-Specific Conclusions

9.18 Selected Approach and Rationale

The regional boundary treatment is selected at specification level. Regional-to-local transfer remains an interface workstream: the required quantities are identified, but final projection operators, synchronised coupling residuals and acceptance tolerances are not yet selected.

The local boundary treatment is selected at role level: inlet forcing from regional histories, open/radiation downstream and lateral boundaries, open atmospheric top and wall-function-compatible no-penetration/no-slip treatment on terrain and barrier patches. Exact relaxation profiles, pressure-reference handling, wall functions and roughness values remain numerical and validation decisions.

9.19 Unresolved Decisions

Open items are final artificial-boundary placement, domain-specific sponge settings, source/bathymetry projection after data lock, and regional-to-local coupling operators. Local open items are inlet relaxation strategy, side/outlet absorption coefficients, pressure reference treatment, wall-function constants, roughness parameters and coupling-interface residuals.

9.20 Requirements and Handoffs

The Software boundary implementation shall expose boundary-face state construction for open/radiation, sponge, reflective and transmissive states. Verification shall test boundary reflection, sponge damping, wet/dry coastline behaviour and interface transfer separately.

For the local model, Software shall expose boundary-face state construction for inlet histories, outlet/side radiation or relaxation, open top and terrain/barrier wall patches. Verification shall test boundary reflection and inlet reconstruction before local force or overtopping results are accepted.

9.21 Deliverable Completion Record

10 RES-NUM-STB — Stability, Consistency and Convergence Analysis

10.1 Purpose and Scope

This Deliverable records stability, boundedness and convergence requirements for the local three-dimensional URANS–VOF finite-volume model. It defines numerical acceptance checks; it does not replace physical validation against observations or experiments.

10.2 Research Questions

10.3 Terminology and Definitions

10.4 Literature Evidence

Hirt and Nichols [5], Roenby, Bredmose, and Jasak [7] and Vukčević et al. [6] are retained for VOF boundedness, interface sharpness and finite-volume phase conservation. Source role: ‘3D-VOF’. Issa [8] is retained for pressure correction and residual convergence in segregated incompressible solves. Source role: ‘3D-FVM’. Project conclusion: a local timestep shall be accepted only when phase boundedness, mass conservation, pressure correction and turbulence-variable admissibility are all within tolerance. Limitation: exact tolerances remain validation and implementation parameters.

10.5 Governing Relations and Quantitative Definitions

10.6 Discrete Formulation

The local solver is accepted at a time level only after the segregated VOF, momentum, pressure-correction and turbulence solves have produced admissible fields. The discrete accepted state is:

$$\mathcal{Q}_{\text{local}}^{n+1} = \{\alpha^{n+1}, \rho^{n+1}, \mu^{n+1}, \mathbf{u}^{n+1}, p^{n+1}, k^{n+1}, \omega^{n+1}, \mu_t^{n+1}, \phi^{n+1}\} \quad (10.1)$$

10.7 Conservation Properties

The discrete mass residual in a cell is:

$$R_{m,i} = \sum_{f \in \partial K_i} \phi_f \quad (10.2)$$

The normalised mass residual is:

$$\hat{R}_{m,i} = \frac{|R_{m,i}|}{\sum_{f \in \partial K_i} |\phi_f| + \epsilon_\phi} \quad (10.3)$$

The global phase-volume balance shall be reported using:

$$M_\alpha = \sum_i V_i \alpha_i \quad (10.4)$$

10.8 Consistency and Formal Accuracy

The local URANS–VOF branch shall report whether a result used first-order fallback, interface clipping, non-orthogonal correction limits, turbulence clipping or timestep rejection. A result that meets a residual tolerance only after repeated limiting shall not be treated as formally equivalent to an unlimited converged result.

10.9 Stability Requirements

The VOF field shall remain bounded:

$$0 \leq \alpha_i \leq 1 \quad (10.5)$$

Mixture properties shall remain positive:

$$\rho_i > 0 \quad (10.6)$$

$$\mu_i > 0 \quad (10.7)$$

Turbulence variables shall remain admissible:

$$k_i \geq 0 \quad (10.8)$$

$$\omega_i > 0 \quad (10.9)$$

The pressure-correction loop shall reduce $\hat{R}_{m,i}$ below the configured tolerance before a timestep is accepted.

10.10 Mesh and Time-Step Requirements

The timestep shall satisfy the local constraints in Equation (7.1). A timestep may be rejected after the solve when any boundedness, residual or load-spike acceptance criterion fails.

10.11 Numerical Failure Modes

Local numerical failure modes include:

- loss of VOF boundedness;
- negative density, viscosity, k or ω ;
- pressure–velocity decoupling on a collocated mesh;
- failure of the pressure-correction loop to reduce mass residuals;
- excessive non-orthogonal correction or wall-function violation;
- artificial boundary reflection contaminating the impact region;
- nonphysical pressure or force spikes caused by interface smearing, timestep size or mesh quality;
- inconsistent force and moment integrals on the barrier patch.

10.12 Verification Requirements

Verification shall include timestep-refinement and pressure-corrector studies for:

- phase-volume conservation;

-
- free-surface boundedness and sharpness;
 - cellwise and global mass residuals;
 - pressure and velocity convergence;
 - k and ω boundedness;
 - force and moment sensitivity to timestep and pressure residual tolerance.

10.13 Candidate Approaches

10.14 Critical Comparison

10.15 Assumptions and Applicability

10.16 Limitations and Failure Conditions

10.17 Project-Specific Conclusions

10.18 Selected Approach and Rationale

The selected local stability policy is accept-or-reject timestep control. A timestep is accepted only when VOF boundedness, physical property positivity, turbulence-variable admissibility, pressure/mass convergence and output-quality checks all pass. Otherwise the timestep is repeated with a reduced Δt or the simulation is marked failed according to the configured recovery limit.

10.19 Unresolved Decisions

Open items are the final residual tolerances, maximum pressure corrector count, non-orthogonal correction limit, local timestep reduction factor, maximum rejected-timestep count and load-spike diagnostic threshold.

10.20 Requirements and Handoffs

‘RES-NUM-SPEC’ shall call these checks before accepting a local timestep. ‘RES-VER-MET’ shall distinguish numerical acceptance from physical validation, especially for pressure, force, moment and overtopping outputs.

10.21 Deliverable Completion Record

11 RES-NUM-ERR — Numerical Error, Adaptivity and Quality Indicators

11.1 Purpose and Scope

This Deliverable defines numerical diagnostics for the local three-dimensional URANS–VOF model. It records what must be measured to judge conservation, mesh quality, refinement needs and output reliability. It does not set final validation tolerances.

11.2 Research Questions

11.3 Terminology and Definitions

11.4 Literature Evidence

Roenby, Bredmose, and Jasak [7] and Vukčević et al. [6] are retained as evidence that VOF quality must be assessed through phase conservation, boundedness and interface sharpness. Source role: ‘3D-VOF’. Project conclusion: local diagnostic output shall include both conservation quantities and engineering-output quality indicators. Limitation: these diagnostics support numerical quality; they do not replace physical validation against benchmark or experimental data.

11.5 Governing Relations and Quantitative Definitions

The global VOF-volume error over a timestep is:

$$E_{\alpha}^{n+1} = \frac{M_{\alpha}^{n+1} - M_{\alpha}^n + \Delta t \sum_{f \in \partial\Omega} \phi_{\alpha,f}}{M_{\alpha}^n + \epsilon_{\alpha}} \quad (11.1)$$

The global mass imbalance is:

$$E_m = \frac{|\sum_i R_{m,i}|}{\sum_i \sum_{f \in \partial K_i} |\phi_f| + \epsilon_{\phi}} \quad (11.2)$$

11.6 Discrete Formulation

Each accepted local timestep shall write a numerical-quality record containing timestep size, Courant numbers, VOF boundedness corrections, pressure residuals, mass residuals, turbulence residuals, mesh-quality extrema and output-integral consistency checks.

11.7 Conservation Properties

The conservation diagnostics shall include:

- global and local phase-volume balance;
- cellwise and global mass residuals;
- net boundary fluxes through inlet, outlet, side and top patches;
- momentum-flux balance across regional-to-local interfaces;
- force and moment balance on terrain and barrier patches where those quantities are extracted.

11.8 Consistency and Formal Accuracy

11.9 Stability Requirements

11.10 Mesh and Time-Step Requirements

The local refinement indicators shall include:

- $\|\nabla\alpha\|$ near the air–water interface;
- local Courant and interface Courant numbers;
- pressure gradient and pressure impulse near the barrier;
- vorticity and turbulent-viscosity ratio in wakes and separation regions;
- wall-distance and wall-function validity indicators;
- force and moment sensitivity to mesh refinement;
- overtopping-volume sensitivity to crest and run-up resolution.

11.11 Numerical Failure Modes

A simulation quality record shall flag:

- clipped phase fractions or turbulence variables;
- repeated timestep rejection;
- pressure residual stagnation;
- large global mass imbalance;
- local mesh-quality excursions in impact or wall regions;

-
- abrupt pressure, force or moment spikes not supported by neighbouring mesh or timestep refinements;
 - boundary-reflection indicators above tolerance.

11.12 Verification Requirements

Verification shall demonstrate that free-surface elevation, overtopping volume, peak pressure, resultant force, impulse and moment converge under mesh and timestep refinement before the local model is used to compare candidate barriers.

11.13 Candidate Approaches

11.14 Critical Comparison

11.15 Assumptions and Applicability

11.16 Limitations and Failure Conditions

11.17 Project-Specific Conclusions

11.18 Selected Approach and Rationale

The selected diagnostic approach is to report numerical quality alongside every local engineering output. A force, moment or overtopping value is not accepted as a standalone scalar unless the supporting conservation, boundedness, mesh and timestep diagnostics are available.

11.19 Unresolved Decisions

Open diagnostic decisions are final numerical-quality thresholds, adaptive-refinement triggers, output smoothing rules for pressure and force histories, and the minimum convergence rate required between mesh levels.

11.20 Requirements and Handoffs

Software shall emit the diagnostics listed in this Deliverable for every local simulation used in validation, surrogate training or candidate comparison. ‘RES-VER-MET’ shall define which diagnostic failures invalidate a physical validation comparison.

11.21 Deliverable Completion Record

12 RES-NUM-SPEC — Integrated Numerical-Method Specification

12.1 Purpose and Scope

This Deliverable records the integrated numerical decision status across the regional hydrodynamic solver, local three-dimensional CFD solver, structural solver, two-dimensional–three-dimensional coupling solver, and FSI coupling solver.

It does not mark the whole numerical framework as complete. The regional nonlinear shallow-water finite-volume branch is now complete at research-specification level. The local three-dimensional URANS–VOF hydrodynamic branch is now specified at solving-procedure level. Structural dynamics, regional-to-local transfer details, FSI coupling, implementation and validation remain separate workstreams.

12.2 Research Questions

The current phase records:

1. which solver-family decisions are complete;
2. which lower-level regional and local choices remain provisional or unresolved;
3. which structural and coupling solver choices remain open;
4. which software architecture references are selected without constraining every solver to one implementation.

12.3 Terminology and Definitions

The decision statuses are:

selected The project adopts the item for the current branch.

provisional The item is the current working direction but still requires lower-level comparison or verification.

excluded The item is removed from the active branch.

unresolved The item remains open and requires later evidence.

12.4 Breadth-First Numerical Research Strategy

The project adopts a breadth-first numerical research strategy:

define the governing models → select a numerical solver family for each model →
select numerical coupling methods → return to shared framework components →
increase detail within each selected branch

The current solver-family order is:

1. regional NLSWE solver;
2. local three-dimensional two-phase fluid solver;
3. structural dynamics solver;
4. two-dimensional–three-dimensional coupling solver;
5. fluid–structure coupling solver.

After these family-level decisions are established, the research shall return to shared numerical components: initial conditions, boundary conditions, source-term discretisation, mesh and refinement, numerical fluxes, reconstruction, time integration, linear and nonlinear solvers, coupling residuals, convergence criteria, verification, and validation.

The finite-volume selection in ‘RES-NUM-FRM’ completes the regional solver-family decision. The local three-dimensional selection in Section [12.20.1](#) completes the local fluid-solver family decision while leaving lower-level local algorithms provisional or unresolved.

12.5 Numerical Decision Status

12.6 Literature Evidence

12.7 Governing Relations and Quantitative Definitions

12.8 Discrete Formulation

The integrated numerical framework shall permit solver families to differ across governing systems. The regional solver may use a standalone finite-volume implementation, while the local three-dimensional CFD solver may use OpenFOAM as software and architecture reference.

The regional implementation shall not be forced inside OpenFOAM. Shared abstractions shall be transferred where they improve consistency across solvers.

Item	Status	Decision statement
Regional governing model	selected	Two-dimensional nonlinear shallow-water equations with variable bathymetry, source forcing, wetting and drying, and optional later dispersive extension
Regional numerical family	selected	Cell-centred finite-volume method, as recorded in Section 3.18.1
Regional solver composition	selected	Polygonal finite-volume update with triangular and quadrilateral practical meshes, MUSCL reconstruction, weighted least-squares gradients, Barth–Jespersen limiter, hydrostatic reconstruction, HLLC primary flux with HLL/HLLC fallback, wet/dry positivity controls, SSPRK(3,3) transport, Strang-split Manning friction, CFL timestep, open/radiation boundaries and optional sponge damping
Finite difference for production regional solver	excluded	Not selected for production, retained as a numerical reference and source of semi-implicit architectural ideas
Lattice/discrete Boltzmann for regional solver	excluded	Removed from the active shortlist pending tsunami validation, dry-front, positivity, well-balancing and complex-bathymetry evidence
Spectral and pseudospectral methods	excluded	Screened out for this project branch owing to irregular coastal boundaries, wetting and drying, bores, discontinuities, bathymetry and local refinement needs
OpenFOAM local CFD reference	selected	Principal open-source software and architectural reference for the local three-dimensional CFD model
Local three-dimensional numerical family	selected	Eulerian cell-centred polyhedral finite-volume method with volume-of-fluid interface capture, pressure-correction or projection incompressibility treatment, URANS $k-\omega$ SST operational closure, and LES retained as a higher-fidelity comparison mode
Local three-dimensional solve procedure	selected	Segregated URANS–VOF update with regional inlet forcing, timestep selection, bounded VOF transport, mixture-property update, momentum predictor, pressure correction, turbulence solve, wall-function update, metric extraction and accept/reject control
Structural solver	unresolved	Element family, constitutive discretisation and structural time integration remain to be selected
Two-dimensional–three-dimensional numerical coupling	unresolved	Transfer operators, coupling residuals and convergence criteria remain to be selected
FSI numerical coupling	unresolved	One-way load transfer is retained and two-way coupling remains conditional on structural feedback; the numerical algorithm is not yet selected

Table 1. Current numerical decision status for the breadth-first research phase

12.8.1 Regional 2D NLSWE Finite-Volume Specification

The regional 2D NLSWE numerical model is established at research-specification level. The solved state is the horizontal two-dimensional conserved vector:

$$\mathbf{U}_i(t) = \frac{1}{A_i} \int_{K_i} \mathbf{U}(\mathbf{x}, t) dA \quad (12.1)$$

For cell K_i with area A_i , boundary faces f , face lengths L_f , outward normals $\mathbf{n}_f$, and source contribution $\mathbf{S}_i$, the finite-volume update is:

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{A_i} \sum_{f \in \partial K_i} L_f \hat{\mathbf{F}}_{if} + \mathbf{S}_i \quad (12.2)$$

For a polygon with N_i faces this is written:

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{A_i} \sum_{k=1}^{N_i} L_{i,k} \hat{\mathbf{F}}_{i,k} + \mathbf{S}_i \quad (12.3)$$

Cell shape changes only the geometry entering the update:

$$A_i, \quad N_i, \quad L_{i,k}, \quad \mathbf{n}_{i,k}, \quad \mathbf{x}_{f_{i,k}}, \quad \mathcal{N}(i) \quad (12.4)$$

Primary practical regional meshes use unstructured triangles. Quads are retained for tests and simple offshore corridors. General straight-edged polygons remain admissible where mesh quality and validation evidence support them.

12.8.1.1 Geometry.

For counter-clockwise vertices $\mathbf{x}_{i,1}, \dots, \mathbf{x}_{i,N_i}$, with $\mathbf{x}_{i,N_i+1} = \mathbf{x}_{i,1}$:

$$L_{i,k} = \|\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}\| \quad (12.5)$$

$$\mathbf{x}_{f_{i,k}} = \frac{\mathbf{x}_{i,k} + \mathbf{x}_{i,k+1}}{2} \quad (12.6)$$

$$\mathbf{t}_{i,k} = \frac{\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}}{L_{i,k}} \quad (12.7)$$

$$\mathbf{n}_{i,k} = [t_y, -t_x]^\top \quad (12.8)$$

For a triangular cell:

$$A_i = \frac{1}{2} |(\mathbf{x}_2 - \mathbf{x}_1) \times (\mathbf{x}_3 - \mathbf{x}_1)| \quad (12.9)$$

$$\mathbf{x}_i = \frac{\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3}{3} \quad (12.10)$$

For a counter-clockwise quadrilateral:

$$J_k = x_k y_{k+1} - x_{k+1} y_k \quad (12.11)$$

$$A_i = \frac{1}{2} \sum_{k=1}^4 J_k \quad (12.12)$$

$$x_i = \frac{1}{6A_i} \sum_{k=1}^4 (x_k + x_{k+1}) J_k \quad (12.13)$$

$$y_i = \frac{1}{6A_i} \sum_{k=1}^4 (y_k + y_{k+1}) J_k \quad (12.14)$$

12.8.1.2 Reconstruction and Flux.

The reconstructed variables are:

$$\mathbf{W} = [\eta, u, v, b]^\top \quad (12.15)$$

For scalar ϕ , the face value is reconstructed as:

$$\phi_{i,f} = \phi_i + \alpha_i^\phi \nabla \phi_i \cdot (\mathbf{x}_f - \mathbf{x}_i) \quad (12.16)$$

Weighted least-squares gradients are selected:

$$\mathbf{d}_{ij} = \mathbf{x}_j - \mathbf{x}_i \quad (12.17)$$

$$\mathbf{M}_i = \sum_{j \in \mathcal{N}(i)} w_{ij} \mathbf{d}_{ij} \mathbf{d}_{ij}^\top \quad (12.18)$$

$$\mathbf{r}_i = \sum_{j \in \mathcal{N}(i)} w_{ij} \mathbf{d}_{ij} (\phi_j - \phi_i) \quad (12.19)$$

$$\nabla \phi_i = \mathbf{M}_i^{-1} \mathbf{r}_i \quad (12.20)$$

The baseline limiter is Barth–Jespersen. Reconstruction is reverted to first order near wet/dry fronts where required for positivity.

HLLC is the primary face flux for wet, valid states. HLL or HLLE is the fallback near dry, degenerate or unsafe states. Each face flux is solved in normal/tangential coordinates:

$$u_n = un_x + vn_y \quad (12.21)$$

$$u_t = -un_y + vn_x \quad (12.22)$$

$$\tilde{\mathbf{U}}_K = [h_K, h_K u_{n,K}, h_K u_{t,K}]^T, \quad K \in \{L, R\} \quad (12.23)$$

12.8.1.3 Hydrostatic Reconstruction and Sources.

Hydrostatic reconstruction uses:

$$b_f^H = \max(b_L, b_R) \quad (12.24)$$

$$h_L^H = \max(0, \eta_L - b_f^H) \quad (12.25)$$

$$h_R^H = \max(0, \eta_R - b_f^H) \quad (12.26)$$

The second-order bed source is:

$$\mathbf{S}_{b,i}^{(2)} = [0, -gh_i(b_x)_i, -gh_i(b_y)_i]^T \quad (12.27)$$

The sponge source uses:

$$\mathbf{S}_{sp,i} = -\sigma_i [h_i - h_{\text{ref},i}, q_{x,i}, q_{y,i}]^T \quad (12.28)$$

$$h_{\text{ref},i} = \max(\eta_{\text{ref}} - b_i, 0) \quad (12.29)$$

$$\sigma_i = \sigma_{\text{max}} \left(\frac{d_i}{L_{\text{sp}}} \right)^p \quad (12.30)$$

$$\sigma_{\text{max}} = \frac{(p+1)c_{\text{ref}}}{L_{\text{sp}}} \ln \left(\frac{1}{A_{\text{target}}} \right) \quad (12.31)$$

The baseline sponge values are $p = 2$, $A_{\text{target}} = 0.01$, and $L_{\text{sp}} = 50\text{--}100$ km before adjustment to the final domain scale.

12.8.1.4 Wetting, Drying and Time Integration.

The regional solver uses positivity-preserving wet/dry treatment, not a regional VOF method. Dry cells are reset by:

$$h_i < h_{\text{dry}} \quad \Rightarrow \quad h_i = 0, \quad q_{x,i} = q_{y,i} = 0 \quad (12.32)$$

The draining limiter is:

$$\lambda_i = \min \left(1, \frac{A_i \max(h_i - h_{\text{dry}}, 0)}{\Delta t \sum_f L_f \max(F_{h,if}, 0) + \varepsilon} \right) \quad (12.33)$$

The transport/friction time update is:

$$\mathbf{U}^{n+1} = \mathcal{M}_{\Delta t/2} \circ \mathcal{T}_{\Delta t}^{\text{SSPRK3}} \circ \mathcal{M}_{\Delta t/2} (\mathbf{U}^n) \quad (12.34)$$

The CFL timestep is:

$$a_f = \max_K \left(|u_{n,K}| + \sqrt{gh_K} \right) \quad (12.35)$$

$$\Delta t_i = C_{\text{CFL}} \frac{A_i}{\sum_{f \in \partial K_i} L_f a_f} \quad (12.36)$$

$$\Delta t = \min_i \Delta t_i \quad (12.37)$$

12.8.1.5 Established and Deferred Scope.

Established for the regional model	Deferred or separate
Regional 2D NLSWE finite-volume formulation; polygonal geometry; triangle/quadrilateral practical meshes; reconstruction and limiter; hydrostatic reconstruction; HLLC/HLL flux strategy; wetting/drying; dynamic bed treatment; SSPRK(3,3) transport with split Manning; CFL timestep; open/radiation and sponge treatment; mesh-refinement principles	QGIS/geospatial domain lock; exact bathymetry/topography dataset; exact fault-source dataset; validation runs; local 3D/interface/FSI model; production software implementation; optimisation and ML workflow

Table 2. Regional numerical-model established/deferred status

12.9 Conservation Properties

12.10 Consistency and Formal Accuracy

12.11 Stability Requirements

12.12 Mesh and Time-Step Requirements

12.13 Numerical Failure Modes

12.14 Verification Requirements

12.15 Candidate Approaches

The active breadth-first branches are:

-
- regional NLSWE finite-volume solver;
 - local three-dimensional two-phase CFD solver;
 - structural dynamics solver;
 - two-dimensional–three-dimensional coupling solver;
 - fluid–structure coupling solver.

12.16 Critical Comparison

12.17 Assumptions and Applicability

12.18 Limitations and Failure Conditions

12.19 Project-Specific Conclusions

The project has selected the regional finite-volume family, completed the regional 2D NLSWE numerical model at research-specification level, and selected the local three-dimensional two-phase finite-volume solver family. It has not completed structural, dimensional-coupling or FSI numerical-method selection. The numerical architecture shall remain modular while implementation and validation work proceeds.

12.20 Selected Approach and Rationale

12.20.1 Selection of the Local Three-Dimensional Fluid Solver

The selected numerical framework for the local three-dimensional fluid model is an Eulerian, cell-centred, polyhedral finite-volume method combined with a volume-of-fluid interface-capturing method and a pressure-correction procedure. This framework will discretise the incompressible, immiscible, two-phase Navier–Stokes equations. The same fluid-solver architecture will support either unsteady Reynolds-averaged Navier–Stokes or large-eddy simulation turbulence treatment, with URANS $k-\omega$ SST selected as the operational baseline and LES retained for selected high-fidelity comparison cases.

The numerical-family decision is therefore:

cell-centred polyhedral finite volume + volume-of-fluid interface capture +
pressure correction

This decision fixes the principal spatial, interface and incompressibility enforcement families. It does not yet fix the exact volume-of-fluid algorithm, pressure-correction algorithm, temporal discretisation, wall-function constants, mesh-motion method or fluid–structure coupling procedure.

12.20.1.1 Conservative Fluid and Phase Transport.

The finite-volume method evaluates the change in each control-volume state through fluxes crossing its bounding faces. This provides a direct conservative representation of fluid transport and supplies a consistent basis for calculating momentum exchange and structural loading. A common face flux also enables neighbouring control volumes to exchange equal and opposite contributions, which is advantageous for global conservation auditing and domain decomposition.

The volume-of-fluid method was selected as the free-surface representation since it tracks the volume fraction of each phase and is therefore naturally compatible with conservative finite-volume transport. Geometric volume-of-fluid methods have been developed for arbitrary polyhedral meshes and have demonstrated strong phase-volume conservation, boundedness and interface sharpness [7].

12.20.1.2 Breaking, Overtopping and Interface Topology.

The local solver must represent wave breaking, overturning, overtopping, jet formation, fragmentation and interface reconnection. An Eulerian interface-capturing method allows these changes in topology without requiring the computational mesh to conform to the air–water interface.

A sharp finite-volume free-surface formulation combining geometric volume-of-fluid advection with a Ghost Fluid treatment has been verified for gravity-wave propagation and validated for violent green-water flow [6]. This provides direct methodological evidence that the selected family can represent both smooth wave propagation and strongly deformed free-surface flow within one solver architecture.

12.20.1.3 Pressure and Structural-Load Prediction.

The structural analysis requires reliable predictions of local pressure, pressure impulse, resultant force, shear traction and overturning moment. The pressure-correction method enforces the incompressibility constraint by coupling the momentum predictor to a pressure equation and subsequently correcting the velocity and face-flux fields.

The large density change across the air–water interface requires particular care. A diffuse pressure and density transition may alter the magnitude and timing of an impact load. Sharp-interface finite-volume formulations provide a route for imposing the pressure-gradient and density discontinuities consistently while retaining a compact stencil on general cell shapes [6].

12.20.1.4 Mesh and Geometric Flexibility.

Polyhedral finite-volume meshes permit body-fitted representation of irregular barriers, coastal topography and local three-dimensional structures. They also allow the mesh resolution to be concentrated around the free surface, barrier edges, overtopping regions and wakes without imposing a uniformly fine Cartesian grid.

Geometric volume-of-fluid methods have been formulated for structured and unstructured polyhedral meshes [7]. A related OpenFOAM implementation has demonstrated unstructured geometric volume of fluid with local dynamic adaptive mesh refinement [10]. The exact refinement algorithm remains unresolved and will be selected during the detailed mesh-framework study.

12.20.1.5 URANS and LES Compatibility.

Unsteady Reynolds-averaged Navier–Stokes and large-eddy simulation will be treated as alternative operating modes of the same local three-dimensional solver. Both modes will use the same continuity, momentum, phase-fraction, pressure-correction, boundary and force-integration infrastructure.

The unsteady Reynolds-averaged Navier–Stokes mode will provide the operational framework for repeated barrier assessment and optimisation. The large-eddy simulation mode will provide higher-fidelity analysis of selected impact cases in which resolved separation, breaking-generated turbulence, fluctuating pressure or load spectra are important. The baseline Reynolds-averaged closure is k – ω SST [15]. The LES subgrid-scale closure, rough-wall treatment and wall-function constants remain unresolved.

12.20.1.6 Local URANS–VOF Solving Procedure.

At accepted time level t^n , the known local fields are:

$$\mathcal{Q}_{\text{local}}^n = \{\alpha^n, \rho^n, \mu^n, \mathbf{u}^n, p^n, k^n, \omega^n, \mu_t^n, \phi^n\} \quad (12.38)$$

The local segregated update is:

0. Interpolate the accepted regional two-dimensional forcing to the local three-dimensional inlet using the selected regional-to-local history and the boundary reconstruction in ‘RES-MOD-IBC’.
1. Select the timestep from the flow-CFL, interface-CFL, gravity-wave and diffusion constraints in Equation (7.1).
2. Update all boundary-face states for α , ρ , μ , $\mathbf{u}$, p , k , ω and ϕ .
3. Solve the bounded compressive VOF equation for α^{n+1} .
4. Update ρ^{n+1} and μ^{n+1} from α^{n+1} using the mixture-property definitions in ‘RES-MOD-EQS’.
5. Assemble and solve the URANS momentum predictor for $\mathbf{u}^*$.
6. Construct pressure-consistent face fluxes ϕ^* using Rhie–Chow or an equivalent collocated-mesh interpolation.
7. Solve the pressure-correction equation for p' .
8. Correct pressure, velocity and face fluxes.
9. Repeat pressure correctors until the mass residual in Equation (10.2) is acceptable.
10. Solve the k and ω equations using the corrected velocity and corrected face flux.
11. Update μ_t^{n+1} and μ_{eff}^{n+1} .
12. Apply scalable wall-function quantities on terrain and barrier wall patches.

13. Extract pressure, shear, force, moment, run-up, overtopping, transmission and energy metrics.
14. Check boundedness, residual, conservation, mesh-quality and output-quality acceptance criteria.
15. Accept the fields and advance to the next timestep, or reject the timestep and repeat with a reduced Δt .

For any local transported equation with a linearised source $S_{\psi,i} = S_{C,i} + S_{P,i}\psi_i$, the coefficient updates are:

$$a_{P,i}^{\text{new}} = a_{P,i}^{\text{base}} - V_i S_{P,i} \quad (12.39)$$

$$b_i^{\text{new}} = b_i^{\text{base}} + V_i S_{C,i} \quad (12.40)$$

Pressure correction follows the operator-splitting principle of Issa [8]. VOF transport follows the source role ‘3D-VOF’, with Hirt and Nichols [5], Roenby, Bredmose, and Jasak [7] and Vukčević et al. [6] supporting the phase-fraction and sharp-interface requirements. Wave-boundary state updates use source role ‘3D-WAVEBC’, supported by Higuera, Lara, and Losada [13] and Jacobsen, Fuhrman, and Fredsoe [14]. Project-specific conclusion: the solving procedure is event-independent; event-specific inputs change the inlet history, terrain, bathymetry and geometry data, not the algorithmic ordering.

12.20.1.7 Compatibility with the Project Software Architecture.

The selected framework is directly aligned with the decision to use OpenFOAM as the principal open-source software and architectural reference for the local three-dimensional model. The relevant shared abstractions include control volumes, owner–neighbour face connectivity, face-normal fluxes, boundary patches, field operations, pressure–velocity coupling, two-phase interface transport, turbulence-model modularity, domain decomposition, solver dictionaries, runtime configuration and source-level extensibility.

This alignment does not require the regional and local solvers to share one executable. It provides a consistent numerical vocabulary across the regional finite-volume solver, the local OpenFOAM-informed solver and the two-dimensional–three-dimensional coupling interface. The finite-volume method remains selected on physical and numerical grounds; software compatibility provides an additional implementation advantage rather than the primary justification.

12.20.1.8 Assessment of Alternative Numerical Families.

Stabilised finite elements remain a credible alternative for complex geometry, variational multiscale turbulence modelling and monolithic fluid–structure interaction. A stabilised finite-element tsunami model using tetrahedral meshes, volume-of-fluid interface capture and SUPG/PSPG stabilisation has demonstrated two-dimensional–three-dimensional tsunami coupling [11]. However, adopting this route would require a separate finite-element fluid framework and additional treatment to obtain the local face-conservative structure required by the present architecture.

Structured staggered-grid projection methods provide strong pressure–velocity coupling and efficient regular-grid operators. Their use for the present barrier problem would require

immersed-boundary or cut-cell techniques for irregular geometry, together with a separate Cartesian solver infrastructure.

Particle methods provide natural treatment of severe free-surface deformation. However, pressure reconstruction, wall-traction transfer, explicit representation of the air phase and systematic URANS–LES operation remain less mature for the present requirements. The modified moving-particle semi-implicit method reviewed in this study required changes to the kernel, pressure-Poisson source and free-surface detection to reduce pressure fluctuation sufficiently for tsunami–structure comparison [16].

The alternative methods will therefore remain in the research register as validation references and possible specialist methods, rather than as the production local-fluid solver.

12.20.1.9 Decision Status and Remaining Numerical Choices.

Table 3. Decision status for the local three-dimensional fluid solver

Status	Component	Current decision
Selected	Spatial discretisation	Eulerian, cell-centred polyhedral finite-volume method
Selected	Free-surface family	Volume-of-fluid interface capture
Selected	Incompressibility treatment	Pressure-correction or projection family
Selected	Turbulence architecture	Common solver supporting URANS $k-\omega$ SST for operational assessment and LES for selected comparison cases
Selected	Software direction	OpenFOAM as the principal open-source implementation and architectural reference
Provisional	Interface advection	Geometric volume-of-fluid method, with isoAdvector as the leading candidate
Provisional	Interface pressure treatment	Sharp Ghost Fluid treatment rather than a purely diffuse mixture treatment
Unresolved	Pressure correction	Exact PISO, PIMPLE or related correction sequence
Unresolved	Moving structural boundary	Fixed terrain and fixed barrier patches are selected for the hydrodynamic baseline; mesh motion, immersed boundary, cut cell or overset treatment remains open only for activated FSI
Unresolved	LES closure and wall treatment	LES sub-grid model, wall-function constants and rough-wall treatment
Unresolved	Discretisation details	Temporal scheme, momentum convection, interface compression, linear solvers and preconditioners
Unresolved	Mesh resolution	Static refinement, adaptive refinement and interface-focused mesh criteria

The next breadth-first task is to select the numerical family for the structural equations of

motion. Detailed choices within the local three-dimensional fluid solver will be revisited after the structural solver and the principal coupling frameworks have been selected.

12.21 Unresolved Decisions

The unresolved scientific and numerical decisions are:

- final regional dry-depth tolerance and velocity regularisation constants;
- final regional HLL/HLLC wave-speed safeguards;
- final regional mesh levels and domain-specific sponge placement;
- exact local interface-advection algorithm;
- exact local pressure-correction sequence;
- exact local LES closure, wall-function constants and rough-wall treatment;
- local structural-boundary representation;
- local mesh-resolution and refinement criteria;
- structural element and time-integration method;
- numerical FSI coupling algorithm;
- numerical two-dimensional–three-dimensional transfer operators.

12.22 Requirements and Handoffs

The next research pass shall close the following non-regional-model items:

1. QGIS/geospatial domain and dataset lock;
2. regional verification and validation test execution;
3. structural dynamics solver;
4. two-dimensional–three-dimensional coupling solver;
5. FSI coupling solver;
6. software implementation details for the locked regional and local hydrodynamic specifications.

Software work may begin with modular data structures and scaffolds: finite-volume mesh and field containers, manifest-driven I/O, boundary-face state containers, regional 2D scaffold, replay-boundary format and local 3D scaffold. This Research deliverable does not authorise implementation of the local URANS–VOF solver before the remaining numerical, validation and data-interface decisions are accepted.

12.23 Deliverable Completion Record

Regional 2D NLSWE and local 3D URANS–VOF numerical-model status: research specification complete; implementation and validation pending.

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